MIL-P-46640 (Ord)
9 March 1962

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M23A2
LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling, and packing for one type of percussion primer designated as Percussion Primer M23A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

JAN-P-223 - Powder, Black
MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-P-20449 - Primer, Percussion, M61, Loading, Assembling, and Packing
MIL-I-45208 - Inspection Requirements, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Inspection Terms and Definitions

FSC: 1390

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DRAWINGS

ORDNANCE CORPS

8802117 - Fixture, Testing, Primer Percussion, Sensitivity Assembly
8831161 - Primer, Percussion, M23A2 Assembly
8833506 - Box, Packing, Ammunition, for Primer Percussion M23A2
8833507 - Box, Set-up, Packing, Ammunition for Primer, Percussion M23A2

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes
IEL-8831161 - Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials.-Materials and parts shall be in accordance with applicable drawings.

3.2 Primer assembly.-The primer assembly shall comply with all requirements specified on Drawing (dwg.) 8831161 and with all requirements specified in applicable specifications.

3.3 Functioning.

3.3.1 Primer assembly less black powder charge.

3.3.1.1 Sensitivity.-The primer assembly shall not function by an applied energy of 47.52 inch ounces minimum (min.) when tested as specified in 4.3.1.1.1.

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3.3.1.2 Firing.-The primer assembly shall function by an applied energy of 159.20 inch ounces maximum (max.) when tested as specified in 4.3.1.1.2.

3.3.2 Primer assembly firing.-The primer assembly shall function without evidence of hangfire (see 6.2), squib (see 6.3), rupture, or detached parts when tested as specified in 4.3.1.2.

3.4 Moisture content of black powder.-The moisture content of the black powder at the loading station at the time of loading shall not exceed 0.5 percent when tested as specified in 4.3.2.

3.5 Workmanship.-All parts shall be dry and free of burrs, chips, dirt, grease, cleaning agent residue and other foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208, except that the height of drop and weight of ball for the sensitivity and firing tests shall not vary from the procedure specified in 4.3.

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4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

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4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of percussion primer assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number. Each lot shall contain:

a. Components of one interfix lot number from one manufacturer.

b. Black powder from only one interfix lot number from one manufacturer.

c. M61 percussion primer (hereinafter referred to as percussion element) of one interfix lot number from one manufacturer.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Assembly, prior to assembling battery cup (see dwg. 8831161)

Categories	Defects	Method of Inspection	Code No 01 (see 6.4)
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Flash hole obstructed	Visual	01001
102.	Seal missing or incomplete	Visual	01002
103.	Percussion element missing, inverted, cocked or damaged	Visual	01003
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	01004

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4.2.2.2 Assembly, prior to pouring wax (see dwg. 8831161)

Categories	Defects	Method of Inspection	Code No 02
Critical:			
1.	Firing plug or battery cup above flush	Straight edge	02001
Major:	AQL 0.25 percent		
101.	Weight of charge	Balance	02002
102.	Firing plug or battery cup missing or inverted	Visual	02003
Minor:	AQL 1.00 percent		
201.	Closing disc missing	Visual	02004
202.	Evidence of poor workmanship (see 3.5)	Visual	02005

4.2.2.3 Assembly (see dwg. 8831161)

Categories	Defects	Method of Inspection	Code No 03
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Diameter of flange, max	Gage	03001
102.	Diameter below flange, max	Gage	03002
103.	Wax missing, loose, or quantity insufficient	Visual	03003
Minor:	AQL 1.50 percent		
201.	Length, max	Gage	03004
202.	Wax above flush	Straight edge	03005
203.	Marking misleading or unidentifiable	Visual	03006
204.	Evidence of poor workmanship (see 3.5)	Visual	03007

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4.2.2.4 Packed carton, prior to sealing (see dwg. 8833507)

Categories	Defects	Method of Inspection	Code No 04
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Any packing material missing...	Visual	04001
202.	Separator damaged to extent that primers touch each other..	Visual	04002
203.	Primer missing.....	Visual	04003

4.2.2.5 Sealed packing carton (see dwg. 8833507)

Categories	Defects	Method of Inspection	Code No 05
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Carton cut, punctured, or torn.	Visual	05001
102.	Sealing strip incomplete, badly wrinkled, torn, or otherwise fails to seal carton.....	Visual	05002
Minor:	AQL 1.00 percent		
201.	Contents loose.....	Manual	05003
202.	Marking misleading or unidentifiable.....	Visual	05004

4.2.2.6 Sealed barrier bag, when applicable (see dwg. 8833507)

Categories	Defects	Method of Inspection	Code No 06
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Seal mislocated or incomplete..	Visual	06001
102.	Bag torn, cut, or punctured....	Visual	06002
Minor: None defined			

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4.2.2.7 Packing box, prior to sealing (see dwg. 8833506)

Categories	Defects	Method of Inspection	Code No 07
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Carton missing	Visual	07001
Minor: None defined			

4.2.2.8 Sealed packing box (see dwg. 8833506)

Categories	Defects	Method of Inspection	Code No 08
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Box damaged to extent that contents are exposed or liable to become exposed	Visual	08001
102.	Hardware or strapping missing, broken, or loose	Visual-Manual	08002
Minor:	AQL 1.50 percent		
201.	Car seal missing, unsealed, or incorrectly positioned	Visual	08003
202.	Contents loose	Manual	08004
203.	Marking misleading or unidentifiable	Visual	08005
204.	Hardware or strapping improperly engaged or mislocated	Visual	08006

4.2.3 Production testing.-The following tests shall be performed using Standard MIL-STD-105 for sampling plan requirements unless otherwise specified.

4.2.3.1 Functioning.

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4.2.3.1.1 Sensitivity of primer assembly less black powder charge (see 3.3.1.1) Critical defect Code No. 02006.-The sampling plan for this test shall be in accordance with Table IVA, Code letter I, and AQL of 0.25 percent. The test shall be performed as specified in 4.3.1.1.1.

4.2.3.1.2 Firing of primer assembly less black powder charge (see 3.3.1.2) Major defect Code No. 02007.-A sample of 125 primers shall be selected from each lot for this test. If 2 or more primers fail to comply with the specified requirement the lot shall be rejected. If only one defective is found, a second sample of 250 primers shall be tested. If the combined number of defectives in both the first and second sample is 2 or more, the lot shall be rejected. The test shall be performed as specified in 4.3.1.1.2.

4.2.3.2 Firing of primer assembly (see 3.3.2) Major defect Code No. 03008.-The sampling plan for this test shall be in accordance with Table IVA, Code letter G, and AQL of 0.65 percent. The test shall be performed as specified in 4.3.1.2.

4.2.3.3 Moisture content of black powder (see 3.4) Major defect Code No 09001.-The contractor shall provide adequate controls to insure compliance with the requirements, and shall test for verification, at least one sample of black powder from each eight hours production of primer assemblies. Composite samples shall not be used. The test shall be performed as specified in 4.3.2.

4.2.3.4 Check test for deterioration of M61 percussion primer (herein referred to as percussion element).-If the total elapsed time between original acceptance of a percussion element lot and the assembly of that lot in the primer exceeds two years, or if the percussion elements have been subjected to adverse conditions however brief, at any time since previous tests, each percussion element lot shall be subjected to an must comply with the check test for possible deterioration functioning and non-functioning requirements of Specification MIL-P-20449, immediately before the percussion elements are assembled into the primers. The test shall be performed by the contractor (see 6.5) on percussion elements selected by the Government inspector at the facility assembling the percussion elements into the primers,

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and shall not constitute a basis of rejection against the contractor who loaded the percussion elements, or the contractor assembling the percussion elements into the primers except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. permitted while the percussion element lots were under the jurisdiction of either contractor.

4.2.4 Inspection equipment.-Index of Inspection Equipment List Number 8831161 identifies available approved inspection gage designs. However, inspection equipment, other than that specified, is not available and shall be designed by the contractor. The contractor may also design alternate inspection equipment in order to facilitate his operations. The designs of all inspection equipment prepared by the contractor for use in acceptance inspection or in support of his approved written inspection plan (see 4.1.1) shall be approved by the Government prior to fabrication or procuring of such equipment (see 6.6). Upon approval, the contractor shall fabricate or procure equipment to the design. All acceptance inspection equipment shall be certified by the Government as complying with the approved design prior to its use. Supply and maintenance of inspection equipment shall be in accordance with Specification MIL-I-45607.

4.3 Test methods and procedures.

4.3.1 Functioning.

4.3.1.1 Primer assembly less black powder charge.

4.3.1.1.1 Sensitivity.-The primer assembly less the black powder charge shall be assembled in the test fixture specified in 4.2.4. A steel ball approximately one and one half inches in diameter and weighing 7.94 plus or minus 0.02 ounces shall be dropped on the firing pin from a height of 6 inches min. The drop height shall be measured as the distance between the bottom of the ball and the top of the firing pin. Any primer assembly that fires shall be classed defective.

4.3.1.1.2 Firing.-The primer assembly less the black powder charge shall be tested as specified in 4.3.1.1.1 except that the drop height shall be increased to 20 inches max. Any primer assembly that fails to fire shall be classed defective.

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4.3.1.2 Primer assembly firing.-The primer assembly shall be assembled in the test apparatus specified in 4.2.4. The firing of the primer assembly shall be observed to determine compliance with the requirements.

4.3.2 Moisture content of black powder.-The moisture content of the black powder shall be determined as specified in Specification JAN-P-223.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level B (with barrier bag).-The primer shall be preserved and packaged in accordance with dwg. 8833507.

5.1.2 Level C (less barrier bag).-The primer shall be preserved and packaged in accordance with dwg. 8833507.

5.2 Packing.

5.2.1 Level B.-The cartons shall be packed in accordance with dwg. 8833506.

5.3 Marking.-Marking shall be as specified on dwgs. 8833506 and 8833507.

5.4 Data cards.-Data card information shall be as specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, level of packing required, and date of this specification.

6.2 Hangfire.-A preceptible delay between initiation and detonation.

6.3 Squib.-A burning of the charge with little or no detonation.

6.4 Inspection code numbers.-The two and five digit code numbers assigned each inspection defect specified herein, are to facilitate future data collection and analysis by the Government.

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6.5 Cost of check test.-The contracting officer will arrange to have the contractor reimbursed for the expense incurred in the performance of the check test.

6.6 Submission of contractor developed inspection equipment designs.-Documentation of contractor developed inspection equipment designs identified with the associated examination required by the specification shall be submitted in duplicate to Picatinny Arsenal, Dover, New Jersey, Attn: ORDBB-N4, for approval. This documentation shall include a complete description of the use and application of all standard devices and set-ups. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissable. However, the Arsenal completion date for design review will be based on the date of the final submission of all designs.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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